Part VII — Fluids · Chapter 39

Viscosity, Drag, Lift, and Turbulence

How real fluids resist motion, generate forces, and transition from smooth to chaotic flow

> *A streamlined racing car cuts through air at 300 km/h while a brick dropped from the same height reaches only a fraction of that speed before stopping accelerating. A ball thrown through water slows almost instantly; the same ball thrown through air travels hundreds of metres. Honey pours slowly while water flows freely. An aircraft wing somehow pushes upward against gravity using only the flow of air. These are not failures of the ideal-fluid models in previous chapters — they are consequences of the properties those models deliberately ignored: internal friction within the fluid, resistance at surfaces, and the breakdown of smooth flow into turbulent chaos. This chapter shows what ideal flow leaves out and how those omissions matter enormously in every real engineering application.*

39.1 Why Real Fluids Differ from Ideal Fluids

Chapters 37 and 38 developed the ideal-fluid model: a fluid that is incompressible, inviscid (frictionless), and undergoes steady, irrotational flow. The Continuity Equation and Bernoulli's Equation both rest on these assumptions. This model is genuinely useful — it correctly predicts how pipes narrow to speed flow, how wings generate pressure differences, and how fluid speed relates to pressure. But it fails badly in several important situations.

Real fluids have viscosity — internal friction between adjacent layers of fluid moving at different speeds. Viscosity is why stirring a cup of tea requires effort, why oil must be pumped through pipes, and why blood must be driven by the heart's pressure against resistance in capillaries. The ideal-fluid model sets viscosity to zero, which means it predicts that a ball falling through water would never slow down and that pumping costs no energy — both obviously wrong.

Real fluids also exert drag on objects moving through them. Drag opposes motion and converts kinetic energy to heat. A bullet slows as it travels; a parachute uses drag deliberately to slow descent. At high velocities or with complex geometries, smooth (laminar) flow breaks down into turbulence — chaotic, swirling, energy-dissipating motion that ideal-fluid models cannot describe at all.

Finally, real fluids generate lift on appropriately shaped surfaces — not just because of pressure differences (which Bernoulli captures), but because of how viscosity and the boundary layer direct flow and create circulation. Understanding real fluid behaviour requires going beyond Bernoulli.

Feature	Ideal Fluid	Real Fluid
Viscosity	Zero (frictionless)	Non-zero; varies by substance
Drag on objects	None	Significant; shape- and speed-dependent
Flow type	Laminar, smooth	Can be laminar or turbulent
Energy dissipation	None (Bernoulli conserved)	Heat generated by viscosity, drag, turbulence
Boundary condition	Fluid can slip at surfaces	No-slip condition: fluid sticks to surfaces
Lift	Via pressure difference only	Via pressure + circulation + boundary layer

Table 39.1 *Comparison of ideal and real fluid behaviour. Most engineering fluids problems require at least one of the real-fluid effects listed here.*

39.2 Viscosity and Internal Fluid Friction

Viscosity (η, eta) is the measure of a fluid's resistance to deformation by shear stress. It arises from internal friction between adjacent fluid layers moving at different velocities. When you stir a viscous fluid, each layer drags on the layers beside it, resisting the relative motion.

Consider two parallel plates separated by a fluid. If the bottom plate is stationary and the top plate moves at velocity v, the fluid between them forms a velocity gradient: the layer touching the moving plate moves at v; the layer touching the stationary plate is at rest (the no-slip condition). The intermediate layers have intermediate velocities. The force required to maintain this velocity gradient is:

$$F = \eta A (v / d)$$

Viscous shear force. F = force (N); η = dynamic viscosity (Pa·s); A = plate area (m²); v = relative velocity of the plates (m/s); d = separation between plates (m). The quantity v/d is the velocity gradient or shear rate.

The SI unit of dynamic viscosity is the pascal-second (Pa·s). Older literature uses the poise (P), where 1 Pa·s = 10 P. Water at 20°C has $\eta \approx 1.0 \times 10^{-3}$ Pa·s (1.0 mPa·s). Honey at room temperature has $\eta \approx$ 2–10 Pa·s, roughly 10 000 times more viscous than water.

Kinematic viscosity (ν, nu) is the ratio of dynamic viscosity to fluid density:

$$\nu = \eta / \rho$$

Kinematic viscosity ν (m²/s) = dynamic viscosity η (Pa·s) ÷ density ρ (kg/m³). Used when density effects are combined with viscosity, as in Reynolds number calculations.

Fluid	Temperature	η (Pa·s)	Relative to water
Air	20°C	1.8×10^{-5}	× 0.018
Water	20°C	1.0×10^{-3}	× 1 (reference)
Water	100°C	2.8×10^{-4}	× 0.28
Blood (whole)	37°C	$2.5–4.0 \times 10^{-3}$	× 3–4
Olive oil	20°C	8.4×10^{-2}	× 84
Glycerol	20°C	1.41	× 1 410
Honey	20°C	2–10	× 2 000–10 000
Engine oil (SAE 30)	40°C	0.1	× 100

Table 39.2 *Dynamic viscosities of common fluids. Viscosity spans many orders of magnitude and is strongly temperature-dependent.*

39.3 Factors Affecting Viscosity

Viscosity is not a fixed property; it changes significantly with temperature and, for gases, with pressure.

Temperature Effects

For liquids, viscosity decreases with increasing temperature. Higher temperature gives molecules more kinetic energy, allowing them to overcome intermolecular attractions and slide past each other more easily. This is why engine oil flows more freely when hot, why honey pours easily in summer but sluggishly in winter, and why hot water cleans more effectively than cold. The temperature dependence of liquid viscosity is roughly exponential.

For gases, the mechanism is different and the effect is opposite: viscosity increases with temperature. In gases, viscosity arises not from intermolecular attraction but from momentum transfer between molecules colliding between adjacent flow layers. Higher temperature means higher molecular speeds and more frequent inter-layer collisions, increasing the effective internal friction. Air at 100°C is more viscous than air at 20°C.

Pressure Effects

For most liquids, viscosity is nearly independent of pressure at pressures encountered in everyday engineering. For gases, viscosity is also nearly independent of pressure at low to moderate pressures (Boyle’s law region), because the increase in molecular density is offset by a decrease in mean free path. At very high pressures (thousands of atmospheres), liquid viscosity increases significantly — relevant in hydraulic machinery and geological systems.

Newtonian vs. Non-Newtonian Fluids

A Newtonian fluid obeys the shear force equation $F = \eta A(v/d)$ with a constant η regardless of the shear rate. Water, air, most oils, and simple gases are Newtonian. A non-Newtonian fluid has an effective viscosity that changes with shear rate: ketchup, blood, paint, and many polymer solutions are non-Newtonian. This chapter deals only with Newtonian fluids.

39.4 Drag Forces in Fluids

Drag is the force exerted by a fluid on an object moving through it, acting in the direction opposing the object's motion. Every object moving through a fluid experiences drag. The drag force depends on the size and shape of the object, its speed, and the properties (density and viscosity) of the fluid.

The general drag equation is:

$$F_D = \tfrac{1}{2} C_D \rho A v^2$$

Drag force equation. F_D = drag force (N); C_D = drag coefficient (dimensionless, depends on shape and flow regime); ρ = fluid density (kg/m³); A = reference area (m², usually cross-sectional area perpendicular to flow); v = speed of object relative to fluid (m/s).

The v^2 dependence means drag grows rapidly with speed: doubling the speed quadruples the drag force. This is why driving a car at highway speed requires vastly more power than driving at city speed, and why breaking the sound barrier is so challenging. At low speeds, the v^2 dependence weakens and a linear v dependence (Stokes drag) can apply.

Object Shape	Approximate C_D	Notes
Long flat plate (parallel to flow)	0.001–0.005	Very low; used in streamlining
Streamlined airfoil	0.04–0.09	Depends on angle of attack
Smooth sphere	0.47	Classic value; varies with Re
Rough sphere (golf ball)	0.25–0.35	Dimples reduce drag vs smooth sphere
Cylinder (axis perpendicular to flow)	1.0–1.2	Bluff body
Flat plate (perpendicular to flow)	1.28	Maximum drag for flat shape
Cube	1.05	Another bluff body example
Sports car (typical)	0.25–0.35	A major design target
Open parachute	1.3–1.5	Designed for maximum drag

Table 39.3 *Approximate drag coefficients C_D for common shapes. Lower C_D means less drag at the same speed and reference area.*

39.5 Types of Drag

Total drag on an object has several components arising from different physical mechanisms.

Pressure (Form) Drag

Pressure drag arises from the difference in pressure between the front and rear of a moving object. As an object pushes through a fluid, high pressure builds at the stagnation point (the front face) and a region of low pressure forms behind the object (the wake). The net pressure difference pushes back against the motion. Pressure drag dominates for bluff bodies (objects that are not streamlined) and at high Reynolds numbers. A flat plate perpendicular to flow has almost entirely pressure drag.

Skin Friction (Viscous) Drag

Skin friction drag arises from the viscous shear stress at the surface of the object as fluid slides past. The no-slip condition means the fluid immediately at the surface is stationary; just above the surface, the fluid is slow. Viscosity transmits a backward shear force to the object. Skin friction dominates for streamlined bodies (airfoils, fish, submarines) where the shape minimises wake pressure differences. A flat plate parallel to flow has almost entirely skin friction drag.

Induced Drag (Lift-Induced Drag)

Induced drag is the drag that accompanies lift generation on a finite wing. It arises from the wing-tip vortices produced when high-pressure air below the wing spills around the tip to the low-pressure upper surface, creating swirling vortices. These vortices carry energy away from the aircraft and represent a real drag force. Induced drag decreases with speed (unlike form drag) and dominates at low speeds and high angles of attack. It is why aircraft take off at high angle of attack but cruise at lower angles.

Wave Drag

Wave drag occurs near and above the speed of sound. As an aircraft approaches the speed of sound, shockwaves form that carry significant energy, creating a large additional drag component. This is the origin of the “sound barrier”: the steep rise in drag near Mach 1 requires disproportionately large thrust to overcome. Above Mach 1, wave drag decreases but remains significant.

39.6 Drag and Terminal Velocity

When an object falls through a fluid under gravity, drag increases with speed. Initially, gravity exceeds drag and the object accelerates. As speed increases, drag increases until it equals the gravitational force minus any buoyancy. At that point, acceleration is zero and the object falls at constant terminal velocity.

At terminal velocity, net force = 0:

$$mg - F_{buoy} - F_D = 0$$

$$\Rightarrow \quad mg - \rho_{fluid} V g = \tfrac{1}{2} C_D \rho_{fluid} A v_t^2$$

Terminal velocity condition. m = mass of falling object; g = 9.81 m/s²; ρ_fluid = fluid density; V = volume of object (buoyancy term); C_D = drag coefficient; A = cross-sectional area; v_t = terminal velocity.

Solving for terminal velocity (neglecting buoyancy for dense objects in air):

$$v_t = \sqrt{2mg / (C_D \rho_{fluid} A)}$$

Terminal velocity v_t (m/s). Larger and heavier objects have higher terminal velocity; larger cross-section and higher C_D reduce it.

Example — Terminal velocity of a skydiver A skydiver ($m = 80$ kg) in a spread-eagle posture: $C_D \approx 1.0$, $A \approx 0.70$ m², $\rho_{air} = 1.20$ kg/m³. $v_t = \sqrt{2 \times 80 \times 9.81 / (1.0 \times 1.20 \times 0.70)} = \sqrt{1570 / 0.84} = \sqrt{1869} \approx 43$ m/s ≈ 155 km/h. Diving head-first reduces A and C_D, raising v_t to ~90 m/s (320 km/h). A parachute ($C_D \approx 1.4$, $A \approx 50$ m²) reduces v_t to ≈ 5 m/s — safe landing speed.

Example — Terminal velocity of a raindrop A large raindrop: radius 2.0 mm, $m = \rho V = 1000 \times (4/3)\pi(0.002)^3 \approx 3.35 \times 10^{-5}$ kg, $A = \pi(0.002)^2 \approx 1.26 \times 10^{-5}$ m², $C_D \approx 0.47$. $v_t = \sqrt{2 \times 3.35 \times 10^{-5} \times 9.81 / (0.47 \times 1.20 \times 1.26 \times 10^{-5})} \approx 9.6$ m/s. Real large raindrops fall at 7–9 m/s, consistent with this estimate.

Key Point: Terminal velocity is not a fixed value for a given object. It depends on the fluid density (skydivers fall faster at high altitude where air is thinner) and on the object's orientation (a skydiver diving versus spread-eagle). Mass increases terminal velocity; larger cross-section decreases it.

39.7 Lift Forces in Flowing Fluids

Lift is a force exerted by a fluid perpendicular to the direction of relative motion. Unlike drag (which always opposes motion), lift can act in any direction perpendicular to the flow. For aircraft, lift acts upward, supporting the aircraft's weight. For a racing car spoiler, it acts downward, pushing the tyres into the road. For a spinning ball in air, it acts sideways, curving the ball's path.

The lift force is described by the lift equation, analogous to the drag equation:

$$F_L = \frac{1}{2} C_L \rho A v^2$$

Lift force equation. F_L = lift force (N); C_L = lift coefficient (dimensionless); ρ = fluid density (kg/m³); A = reference area (m², typically wing planform area for aircraft); v = airspeed (m/s).

The lift coefficient C_L depends on the shape of the lifting surface and especially on the angle of attack — the angle between the chord line of the wing and the oncoming flow. Increasing angle of attack increases C_L (and lift) up to a critical angle, beyond which the flow separates from the upper surface and C_L drops suddenly. This is called a stall.

39.8 Aerodynamic and Hydrodynamic Lift

Two complementary explanations describe how wings generate lift. Neither is wrong; they are different ways of accounting for the same physical reality.

Pressure Difference Explanation (Bernoulli)

A wing is shaped so that the upper surface is more curved than the lower. Air flowing over the upper surface travels a greater path length in the same time, so it must move faster. By Bernoulli's equation, faster flow corresponds to lower pressure. The higher pressure below and lower pressure above create a net upward force. This explanation is quantitatively correct but leaves out why the air must travel faster over the top — the answer involves the shape and viscosity of the wing and the Kutta condition at the trailing edge.

Momentum Change Explanation (Newton)

A wing deflects air downward. By Newton's third law, if the wing pushes air downward, the air pushes the wing upward. This "downwash" explanation is correct and especially useful for understanding why increasing angle of attack increases lift (more downward deflection) and why lift requires moving air (no deflection in still air relative to the wing).

The Magnus Effect

A spinning ball in a fluid generates lift perpendicular to both its velocity and its spin axis. The spin drags fluid around the ball (via the no-slip condition and viscosity). This creates asymmetric flow: on the side where spin and flow are in the same direction, flow speed is higher and pressure lower; on the other side, speed is lower and pressure higher. The net pressure difference is a sideways force: the Magnus force. This is the origin of the curve ball in baseball, the top-spin dip in tennis, and the banana kick in football (soccer).

> **Example — Aircraft lift** A small aircraft (wing area 16 m^2) flies at 55 m/s. Required lift = weight = 8000 N. Find required C_L: $F_L = \frac{1}{2} C_L \rho A v^2 \Rightarrow C_L = 2F_L / (\rho A v^2) = 2 \times 8000 / (1.20 \times 16 \times 55^2) = 16\,000 / 58\,080 \approx 0.28$. A typical wing can achieve C_L of 1.5–2.0 before stall, so this aircraft has ample margin.

39.9 Boundary Layers

The boundary layer is the thin region of fluid immediately adjacent to a solid surface where the velocity of the fluid transitions from zero (at the surface, due to the no-slip condition) to the free-stream velocity (far from the surface). The no-slip condition is a fundamental result of viscosity: no matter how inviscid a fluid seems, it sticks to any solid surface at zero relative velocity.

The boundary layer can be laminar or turbulent, and this distinction profoundly affects skin friction and the point at which flow separates from the surface. A laminar boundary layer has smooth, layered flow and produces lower skin friction drag. A turbulent boundary layer has chaotic mixing, higher skin friction, but it resists flow separation better than a laminar layer.

Boundary layer separation occurs when the boundary layer detaches from the surface. This typically happens on the rear portion of a curved surface where the pressure is rising (adverse pressure gradient). Separation creates a large turbulent wake with low pressure behind the object — dramatically increasing form drag. Keeping the boundary layer attached as long as possible is a major objective of aerodynamic design: dimples on golf balls trip the boundary layer turbulent early, which keeps it attached longer and reduces total drag compared to a smooth ball at golf-ball Reynolds numbers.

39.10 Laminar vs. Turbulent Flow Revisited

Laminar flow is smooth, organised flow in which fluid moves in parallel layers (laminae) with no lateral mixing between layers. Streamlines do not cross. It is the flow regime assumed throughout Chapters 37–38.

Turbulent flow is chaotic, three-dimensional flow with swirling eddies, rapid velocity fluctuations, and extensive mixing between fluid layers. Most flows in nature and engineering are turbulent: the wake behind a car, smoke rising from a fire, water in a river, and blood in the aorta.

The transition from laminar to turbulent flow is characterised by the Reynolds number (Re), introduced in Chapter 38:

$$\mathrm{Re} = \rho v L / \eta = v L / \nu$$

Reynolds number Re (dimensionless). ρ = fluid density; v = flow speed; L = characteristic length (pipe diameter for pipe flow, chord length for wings); η = dynamic viscosity; ν = kinematic viscosity. Re represents the ratio of inertial forces to viscous forces.

Low Re: viscous forces dominate; flow is laminar and smooth. High Re: inertial forces dominate; flow tends toward turbulence. For pipe flow, the transition typically occurs around Re ≈ 2300. For flow over a flat plate, transition occurs around Re ≈ 5×10^5. These thresholds are not exact — very careful laboratory conditions can maintain laminar flow at Re far above the nominal transition value, but any disturbance triggers turbulence.

39.11 Causes of Turbulence

Turbulence arises when the inertial tendency of fluid parcels to continue moving in their current direction overwhelms the stabilising effect of viscosity, which resists lateral motion between layers. Any perturbation in laminar flow (a bump, a roughness element, a vibration) can grow into turbulence if Re is above the critical value.

Physically, turbulence is sustained by a cycle of vortex stretching: three-dimensional flow structures (vortex tubes) are stretched by the velocity gradient of the mean flow, intensifying their rotation and creating smaller and smaller swirling structures down to the Kolmogorov scale, where viscosity finally dissipates the energy as heat. This energy cascade from large eddies to small eddies is a defining feature of turbulent flow.

Surface roughness promotes turbulence by triggering boundary layer transition earlier and by creating small-scale vortices in the near-wall region. For this reason, aircraft wings and high-performance surfaces are carefully polished. However, as noted for golf balls, early turbulent transition can sometimes be advantageous if it prevents boundary layer separation and reduces form drag.

> **Historical Note:** Osborne Reynolds (1842–1912) performed the classic dye-injection experiment in 1883, demonstrating the laminar-to-turbulent transition in pipe flow and introducing the dimensionless number now bearing his name. He injected a thin stream of coloured dye into the centre of a pipe flow. At low flow speeds the dye remained in a sharp line (laminar); above a critical speed it dispersed throughout the pipe in seconds (turbulent). Reynolds number is now used in virtually every branch of fluid dynamics, aerodynamics, and chemical engineering.

39.12 Engineering Around Fluid Resistance

Engineers use several strategies to manage viscosity, drag, lift, and turbulence in practical applications.

Streamlining

Streamlining reduces pressure drag by shaping objects so that the boundary layer remains attached all the way to the trailing edge, minimising wake size. A teardrop shape is optimal: blunt at the front to resist high pressure, tapered at the rear to guide the flow together without separation. Cars, aircraft, submarines, and fish have all converged on similar streamlined shapes for this reason.

Lubrication

Lubrication uses a viscous fluid (oil, grease) to separate two solid surfaces that would otherwise make direct contact. The fluid layer transmits load between the surfaces through viscous pressure, while preventing the metal-to-metal contact that causes wear. Engine oil viscosity must be high enough to maintain this film but not so high as to require excessive pump energy. This is why multi-grade oils (5W-30, 10W-40) were developed: they behave as thin oils in the cold for easy starting and as thick oils when hot for adequate film strength.

Boundary Layer Control

Aircraft designers use suction slots, vortex generators, and leading-edge modifications to control the boundary layer. Vortex generators are small fins that produce turbulent mixing in the boundary layer, adding energy to the near-wall fluid and delaying separation. They are visible on the wings of many commercial aircraft and on the upper surface of wind turbine blades.

Drag Reduction in Vehicles

Modern cars achieve drag coefficients of 0.25–0.30 through careful shaping of every surface. A 10% reduction in C_D produces roughly a 3–5% improvement in highway fuel economy, because aerodynamic drag dominates at highway speed (it scales as v^2, while rolling resistance is roughly constant). Formula 1 cars use extremely low-drag shapes combined with large aero elements (wings, diffusers) that generate downforce at the cost of increased drag — a deliberate trade-off to improve cornering grip.

39.13 Common Mistakes in Real Fluid Analysis

- Applying Bernoulli where viscosity dominates. Bernoulli's equation assumes inviscid flow. In a situation dominated by viscosity (low Reynolds number, flow through narrow pipes, very viscous fluids), Bernoulli gives incorrect results. Use the viscous flow equation (Poiseuille's Law for pipe flow) or note that energy is being dissipated rather than conserved.
- Forgetting that drag scales as v^2. Because drag grows quadratically with speed, doubling the speed quadruples the drag force and increases the power needed ($F \times v$) by a factor of eight. Many intuitive estimates of drag at high speed are seriously too low.
- Using the wrong reference area in the drag equation. The reference area A in $F_D = \frac{1}{2} C_D \rho A v^2$ is defined as part of the drag coefficient: for spheres and cars it is usually frontal (cross-sectional) area; for wings and flat plates it may be planform (top-view) area. C_D values from tables are meaningless without knowing what reference area was used. Always check the definition when looking up C_D.
- Confusing terminal velocity with final velocity. Terminal velocity is the equilibrium speed at which drag balances gravity. It is not the maximum speed an object can reach under other forces. A powered aircraft can fly far faster than its terminal velocity in free fall. Terminal velocity applies only to free fall in a fluid.
- Assuming laminar flow when Re indicates turbulence. When Reynolds number is above the transition value for the geometry, laminar-flow formulas (Stokes drag $F = 6\pi\eta r v$, Poiseuille's law for pipe flow) no longer apply. Using them above transition gives substantially underestimated drag and pressure drop.
- Neglecting the Magnus effect in ball sports problems. A spinning ball in flight does not follow projectile motion. The Magnus force is comparable to gravity for fast-spinning balls in many sports. Ignoring it gives significantly wrong predicted trajectories.

Common Mistake: The single most common real-fluid error: applying Bernoulli's equation to a flow where viscosity, turbulence, or energy dissipation is significant. Bernoulli is a statement of energy conservation for ideal, inviscid, steady flow. Real-world piping, blood flow, and turbulent aerodynamics all violate one or more of these assumptions.

Chapter Summary

Real fluids differ from ideal fluids through viscosity, drag, lift, and turbulence. Ignoring these effects produces systematically wrong predictions for any engineering system involving fluid flow at significant velocities.

- Viscosity (η) is internal friction between fluid layers; SI unit Pa·s. Liquid viscosity decreases with temperature; gas viscosity increases.
- Drag $F_D = \frac{1}{2}C_D\rho Av^2$ opposes motion; grows as v^2. Components: pressure (form) drag, skin friction, induced drag, wave drag.
- Terminal velocity occurs when drag + buoyancy = weight. $v_t = \sqrt{2mg / (C_D\rho A)}$ for dense objects in air.
- Lift $F_L = \frac{1}{2}C_L\rho Av^2$ acts perpendicular to flow; generated by wings, spinning balls (Magnus effect), and any asymmetric body in flow.
- Boundary layer: thin near-surface region where velocity rises from zero (no-slip) to free-stream. Can be laminar or turbulent; separation causes large form drag.
- Reynolds number $Re = \rho vL/\eta$ predicts laminar ($\lesssim 2300$ for pipes) or turbulent ($\gtrsim 4000$) flow. Turbulence dissipates energy; is caused by inertia overcoming viscosity.
- Engineering strategies: streamlining, lubrication, boundary layer control, vortex generators, optimised drag coefficients.

Key Terms

viscosity (η): The resistance of a fluid to shear deformation; dynamic viscosity in Pa·s; measure of internal fluid friction.

kinematic viscosity (ν): $\nu = \eta/\rho$; ratio of dynamic viscosity to density; units m^2/s.

no-slip condition: The physical condition that fluid in contact with a solid surface has zero velocity relative to that surface; a consequence of viscosity.

drag force (F_D): The fluid resistance force opposing motion; $F_D = ½C_D\rho Av^2$; acts parallel to and opposing the relative velocity.

drag coefficient (C_D): Dimensionless factor characterising the drag of a particular shape in a particular flow regime; defined by $F_D = ½C_D\rho Av^2$.

terminal velocity: The constant velocity of free fall at which drag equals the net downward force (weight minus buoyancy); acceleration is zero.

lift force (F_L): The fluid force perpendicular to the direction of relative motion; $F_L = ½C_L\rho Av^2$.

lift coefficient (C_L): Dimensionless factor characterising the lift of a particular surface shape and angle of attack.

Magnus effect: Lift generated by a spinning object in a fluid flow, due to asymmetric velocity distribution around the spinning body.

boundary layer: The thin region of fluid adjacent to a solid surface where velocity transitions from zero (no-slip) to the free-stream value.

laminar flow: Smooth, orderly flow in parallel layers with no lateral mixing; characterised by low Reynolds number.

turbulent flow: Chaotic, three-dimensional flow with eddies, swirls, and rapid mixing; characterised by high Reynolds number.

Reynolds number (Re): $Re = \rho vL/\eta$; dimensionless ratio of inertial to viscous forces; predicts flow regime (laminar or turbulent).

stall: The sudden loss of lift when angle of attack exceeds the critical value and boundary layer separation occurs over the upper wing surface.

streamlining: Shaping objects to minimise pressure drag by delaying boundary layer separation and reducing wake size.

Concept Review Questions

Answer in complete sentences. No calculations unless stated.

1. Explain why honey is more viscous than water at the molecular level. What happens to honey's viscosity when it is heated?
2. A leaf falls more slowly than a stone of the same mass. Explain using the drag equation and reference area, without invoking density differences.
3. Explain the no-slip condition. Why does this condition lead to a boundary layer? What happens at the outer edge of the boundary layer?
4. A skydiver in a spread-eagle position has a higher terminal velocity than they would wearing a wingsuit. Explain in terms of the drag equation which variable changes and in which direction.
5. Why do golf balls have dimples? What physical effect do the dimples exploit, and what would happen to drag if the dimples were filled in?
6. Explain the two physical explanations for how a wing generates lift. Why are both explanations correct even though they seem to emphasise different things?
7. What is the Magnus effect? Give two examples from ball sports where it plays a significant role.
8. Why does Bernoulli's equation fail to describe flow through a viscous pipe? What has been omitted from the Bernoulli model?

Drag Force Problems

9. A 75 kg cyclist on a road bike (combined frontal area $A = 0.40\ m^2$, $C_D = 0.88$) rides at 10 m/s in still air ($\rho = 1.20\ kg/m^3$). Find the drag force. Then find the drag force at 20 m/s. By what factor does it increase?
10. A car (mass 1200 kg, $A = 2.1\ m^2$, $C_D = 0.28$) travels at 30 m/s. (a) Find the drag force. (b) Find the power needed to overcome drag at this speed. (c) How does power scale with speed?
11. A sphere of diameter 4.0 cm is dropped in air ($\rho = 1.20\ kg/m^3$). C_D for a sphere $= 0.47$. Find the drag force on the sphere when it is moving at 5.0 m/s.
12. A rectangular sign (1.5 m × 0.8 m, flat, perpendicular to wind, $C_D = 1.28$) experiences a wind speed of 25 m/s ($\rho = 1.20\ kg/m^3$). Find the force on the sign mounting. Why might a sign designer deliberately add holes or slits to the sign?

Terminal Velocity Exercises

13. A spherical hailstone (diameter 1.5 cm, density 900 kg/m³, $C_D = 0.47$) falls in air ($\rho = 1.20$ kg/m³). (a) Find the mass and cross-sectional area. (b) Find the terminal velocity. (c) Why might hailstones falling from greater height not necessarily reach higher terminal velocities?
14. A parachutist ($m = 90$ kg) deploys a parachute ($C_D = 1.4$, effective area 55 m²) in air at $\rho = 1.20$ kg/m³. Find the terminal velocity with the parachute open. Is this a safe landing speed (compare to ≈4–5 m/s typical)?
15. Two identical spheres are released simultaneously in two fluids: water ($\rho = 1000$ kg/m³, $\eta = 1.0 \times 10^{-3}$ Pa·s) and glycerol ($\rho = 1260$ kg/m³, $\eta = 1.41$ Pa·s). At low Re, Stokes drag gives $F_D = 6\pi\eta rv$. For a sphere of radius 1.0 mm and density 2500 kg/m³, find the terminal velocity in each fluid. (Use Stokes drag and neglect buoyancy for this estimate, or include it for a better result.)
16. A feather ($m = 0.5$ g, effective area $A = 18$ cm², $C_D = 1.0$) falls in air. Find its terminal velocity. Compare to a steel ball bearing of the same mass (radius ≈ 3 mm, $C_D = 0.47$). What does this show about the role of A and C_D versus mass?

Lift and Aerodynamics Analysis

17. An aircraft wing (planform area 22 m²) must support a total weight of 18 000 N at a cruise speed of 70 m/s in air ($\rho = 1.20$ kg/m³). Find the required lift coefficient C_L.
18. The same aircraft (from Problem 1) has a glide ratio (horizontal distance per vertical drop) of 12:1 at cruise speed. This means $C_L/C_D = 12$. Find C_D and the drag force at cruise speed. What thrust force is needed to maintain level flight?
19. A spinning baseball ($m = 145$ g, radius 3.7 cm) experiences a Magnus lift force of 0.35 N perpendicular to its velocity. Compare this to the ball’s weight. Would this significantly curve the ball’s path over a 18-metre pitch? (You do not need to calculate the trajectory; reason qualitatively.)
20. A car spoiler (inverted wing, $A = 0.60$ m², $C_L = 1.2$) generates downforce at 40 m/s. Find the downforce. If the car has mass 650 kg, what percentage of the car’s weight does the downforce represent? Why is downforce valuable in racing?

Applied Fluid Dynamics Problems

21. Two pipes of equal diameter carry the same fluid. Pipe A has laminar flow (Re = 1200); Pipe B has turbulent flow (Re = 8000). Pipe B requires 3× the pressure difference per unit length to maintain the same average flow speed. Explain physically why turbulent flow requires more driving pressure, and identify where the extra energy goes.
22. A fluid dynamics engineer must choose between two designs for a support strut in airflow: a round cylinder (C_D = 1.2, diameter 4 cm) and a streamlined airfoil shape (C_D = 0.08, same frontal width 4 cm, but depth 18 cm). At 30 m/s airflow (ρ = 1.20 kg/m³) for a 1.0 m long strut, find the drag on each per metre of span. What is the drag reduction ratio? What might be the disadvantage of the streamlined shape?
23. A 200 g tennis ball (radius 3.2 cm, $C_D \approx 0.6$ for rough sphere) is served at 60 m/s. (a) Find the drag force at this speed. (b) Find the deceleration due to drag. (c) How does this compare to gravitational acceleration? What does this tell you about the importance of drag in tennis ball trajectories?
24. Honey (η = 5.0 Pa·s, ρ = 1400 kg/m³) is pumped slowly through a tube of radius 3 mm at average velocity 2.0 cm/s. Calculate Re. Is the flow laminar? Now pump water ($\eta = 1.0 \times 10^{-3}$ Pa·s, ρ = 1000 kg/m³) through the same tube at the same velocity. Calculate Re. Comment on whether Bernoulli's equation is a better approximation for honey or water in this scenario.

Chapter 39 Checkpoint

Before moving on to Chapter 40, confirm you can do each of the following.

I can...	See Section
Explain what an ideal fluid omits and why real fluids differ	39.1
Define viscosity and apply $F = \eta A(v/d)$ to shear flow problems	39.2
Explain how temperature affects viscosity in liquids vs. gases	39.3
Apply the drag equation $F_D = \frac{1}{2}C_D\rho Av^2$ to calculate drag force	39.4
Name and distinguish the four types of drag	39.5
Calculate terminal velocity from force balance	39.6
Apply the lift equation and explain what determines C_L	39.7–39.8
Explain the Magnus effect and give two examples	39.8
Describe the boundary layer and explain the no-slip condition	39.9
Use Reynolds number to predict whether flow is laminar or turbulent	39.10–39.11
Describe at least three engineering strategies to manage fluid resistance	39.12
Identify common errors in applying drag and viscosity concepts	39.13

Continue to Chapter 40: Electric Charge →